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ADVANCING BIOMETRIC AUTHENTICATION: A FRAMEWORK FOR TEMPLATE RECONSTRUCTION ATTACK DETECTION

Project Submitted to the
SRM University AP, Andhra Pradesh
for the partial fulfillment of the requirements to award the degree of

Bachelor of Technology
in
Computer Science & Engineering
School of Engineering & Sciences

submitted by

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May 2026

DECLARATION

I undersigned hereby declare that the project report **Advancing Biometric Authentication: A Framework for Template Reconstruction Attack Detection** submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology in the Computer Science & Engineering, SRM University-AP, is a bonafide work done by me under supervision of Dr.Dilip Kumar Vallabhadas. This submission represents my ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree of any other University.

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CERTIFICATE

This is to certify that the report entitled **Advancing Biometric Authentication: A Framework for Template Reconstruction Attack Detection** submitted by **Koutharapu Babinadh , Y.teja, M. Nithin Sai, D.Satya Koushik** to the SRM University-AP in partial fulfillment of the requirements for the award of the Degree of Master of Technology in in is a bonafide record of the project work carried out under my/our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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Signature: V. Dilip kumar	Signature:

ACKNOWLEDGMENT

***** Given below is an example of acknowledgments. This may be edited. For, example, if you wish, you may thank God for giving you this opportunity to write such a report! ****

I wish to record my indebtedness and thankfulness to all who helped me prepare this Project Report titled **Advancing Biometric Authentication: A Framework for Template Reconstruction Attack Detection** and present it satisfactorily.

I am especially thankful for my guide and supervisor Dr.Dilip Kumar Vallabhadas in the Department of Computer Science & Engineering for giving me valuable suggestions and critical inputs in the preparation of this report. I am also thankful to Dr. Manikandan V M, Head of Department of Computer Science & Engineering for encouragement.

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ABSTRACT

Although fingerprint biometrics are broadly used in authentication systems, they are facing a serious problem: in case the templates are stolen or compromised, the original fingerprint patterns can be restored. Such reconstruction attacks allow adversaries to create synthetic fingerprints to bypass authentication or gain access to large-scale systems. Complementary to most traditional cancelable biometric approaches aiming at template protection, we tackle for the first time the practical problem of detecting such reconstruction attempts before causing misuse. We propose a hybrid dynamic projection-based detection framework identifying inconsistencies between genuine fingerprint embeddings and synthetically reconstructed templates. Based on a ResNet-50 backbone trained with metric learning, our system extracts robust 512-D embeddings and projects them dynamically with a hybrid projection to produce protected templates. The reconstructed samples show measurable deviations in projection consistency, score distribution, and correlation structure, allowing our system to reliably detect attacks while minimally influencing normal authentication accuracy. Based on expensive experiments using the CASIA-FingerprintV5 dataset and extensive simulated attack scenarios, we explain that the proposed approach provides strong attack detection performance at improved high recognition.

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Chapter 1

INTRODUCTION TO THE PROJECT

Biometric authentication [1] is now a cornerstone of modern security systems due to its blend of convenience, user acceptance, and—when properly deployed—high robustness. Fingerprints occupy a particularly dominant position because of their ubiquity across consumer devices and institutional deployments. However, the permanence of biometric traits introduces a key security tension: unlike passwords or tokens, a biometric cannot easily be revoked or replaced if compromised[2]. This immutability amplifies the impact of template leakage. When an attacker obtains a stored template, it can be used as the basis for synthetic biometric generation that replicates the original identity. Recent advances in deep learning, image synthesis, and optimization-based inversion have increased the realism with which protected templates can be inverted or transformed into spoof artifacts. Template reconstruction attacks range from minutiae estimation procedures to fully learned inversion networks (including GAN- and diffusion-based approaches). These attacks do not merely expose a template—they generate usable biometric artifacts that can be replayed against sensors or used to attack other databases (cross-system transfer). Consequently, defenses that only harden templates against inversion may still fail if the attacker succeeds: the reconstructed fingerprint itself can be used in the wild. Motivated by this gap, we redirect attention from passive protection to active detection[3]: rather than only making templates hard-to-invert, we monitor and detect behavioral inconsistencies in protected templates that betray reconstruction attempts. Our key insight is that hybrid, user-specific dynamic projections introduce structural relationships between a user's embedding, its empirical correlation structure, and the randomized key component used to protect the template. Genuine biometric acquisitions preserve these relationships across multiple captures, while reconstructed templates—produced by inversion and synthesis pipelines that do not have access to the true biometric dynamics—exhibit systematic deviations. Detecting these deviations allows operators to flag and stop reconstructed-template use before compromise propagates. The contributions of this paper are threefold: 1. We introduce a hybrid dynamic projection fingerprint behavior model that fuses embeddingdriven correlation structure with user-controlled randomization to produce protected templates whose internal relationships are measurable at authentication time. 2. We design a projection-consistency detection pipeline that computes multiple consistency metrics (correlation preservation, projection variance, score distribution divergence, and embedding reconstruction residuals) and feeds the results to a lightweight classifier to decide whether an input is genuine or reconstructed. 3. We present extensive experiments on CASIA-FingerprintV5 together with simulated and learned inversion attacks, reporting detection and recognition metrics demonstrating the approach's effectiveness and operational feasibility. An important practical requirement for any detection scheme is that it should avoid substantially degrading the primary authentication task. Our experiments show that the proposed method keeps authentication performance within a small, acceptable margin while delivering strong detection results. We also discuss deployment considerations (on-device vs server-side checks), privacy implications, and adaptive attacker scenarios. Additionally, the presented approach has been devised with scalability in mind, which allows it to be easily integrated into an existing authentication system, without making major changes to the architecture thereof. The modularity nature of the suggested framework also makes each component—such as projectors and detection classifier—updateable separately to accommodate any changes that may emerge due to new threat models. Finally, the ability of the framework to adapt to different biometric modalities suggests possible generalization and application to areas other than fingerprint recognition. Future work will revolve around strengthening resilience to more advanced and sophisticated attacks, both adversarial and generative, along with looking into privacy-preserving and federated approaches. By detecting even subtle inconsistencies in the image reconstructed by the attacker, the presented approach allows for increasing the authenticity of authentication systems without compromising their performance. Furthermore, it paves the way towards implementing continuous authentication solutions. With biometric authentication systems becoming more ubiquitous, such techniques will prove crucial to ensuring the integrity and stability of such systems.

Chapter 2

RELATED WORK

Recently, there have been efforts in the area of biometrics security to enhance fingerprint authentication systems against spoofing attacks, replay attacks, and template inversion attacks. Komail Kesana et al. [4]presented a method to combine software-based spoofing detection using CNN, SVM, and ridge frequency and pore analysis to identify spoofed fingerprints created from gelatin or silicone materials. Although this method is useful in defending against spoofing attacks, it is primarily designed to provide sensor-level protection and does not counter template reconstruction attacks. Sani M. Abdullahi et al [5] presented a one-time bio- metric template system that combines deep learning, Bio- Hashing, cryptographic hashing, and symmetric encryption. This method provides an inability for an attacker to defend against replay attacks, but it does not attempt to identify reconstructed templates. Tanguy Gernot and Christophe et al [6] Rosenberger, in their paper "One-time Biometric Template-Based Approach to Prevent Replay Attacks in Traditional Systems," have suggested a one-time biometric template-based approach to prevent replay attacks in traditional systems, which uses deep learning, bio-hashing, and cryptography to provide enhanced security and accuracy in authentication.

Shah Mahmood et al [7]argued that the present condition of biometric template protection techniques is fragmented, either concentrating on standalone cryptographic methods or particular attack models without considering the regulatory and deployment viewpoints. The research work has pointed out the shortcomings in dealing with overall threats like template reversal attacks, cross-match attacks, adversarial attacks, and quantum attacks. To address the above-mentioned shortcomings, the researcher has suggested an attack-centric, cross-modal biometric template protection framework that combines technical, legal, and risk-related viewpoints. The framework has been supplemented by a threat taxonomy and an attack-defense evaluation matrix that help system designers. The framework also suggests a multi-layered protection mechanism that includes cancellable biometrics, cryptographic protection, hardware trust environments, and post-quantum security.

Raffaele Cappelli et al. [8] proposed a method to reconstruct fingerprint images from templates based on minutiae points. The authors proved that realistic images can be reconstructed using mathematical modeling and optimization techniques. The results had a high success rate in breaking automated systems but were not effective in human forensic investigation.

One of the earliest methods in the field of biometric template protection with transformation capabilities to cancel and re-issue compromised templates was introduced by Ratha et al. [9]. This paper established some of the first essential characteristics such as non-invertibility, revocability, and diversity with good matching performance. This paper formed the foundation for all cancelable biometric techniques and also many projection and learning-based techniques.

Abdar et al. [10] have proposed a Binary Residual Feature Fusion (BARF) model for the automated classification of medical images, incorporating Monte Carlo dropout to deal with uncertainties, achieving better results than the existing models. Arora et al. [11] have discussed a deep learning approach to improve the security of biometric systems, reviewing the existing issues and future research trends in biometric safeguarding. Sandhya et al. [12] have proposed a biometric template protection method based on the fingerprint and palmprint representation, where the scores of individual match scores are combined at the score level using T-operators to improve security. Hammad Mohamed et al. [13] have designed a multimodal biometric authentication system using CNN and Q-Gaussian multi-support vector machines (QG-MSVM) with various levels of fusion to protect biometric templates. In addition, Sandhya et al.[14] have proposed a multi-instance cancelable iris authentication system based on convolutional neural networks trained with triplet loss to handle privacy issues in cloud-based biometric substantiation systems. Raffaele Cappelli et al. introduced a new technique for reconstructing fingerprint images from ISO/IEC 19794-2 minutiae templates and showed that standard templates hold enough information to realistically reconstruct fingerprint patterns. Their technique involves the mathematical modeling of fingerprint area, orientation field, and ridge pattern. The reconstructed images were tested against several commercial fingerprint recognition systems to measure susceptibility to masquerade attacks. The research highlights several crucial security concerns with regards to the security of templates in biometric applications. The research study emphasizes the need for incorporating a reliable encryption mechanism along with secure communications into the biometric system. Moreover, the research study suggests the need for incorporating anti-spoofing and liveness detection measures into the design of a biometric system to prevent template reconstruction attacks. Regardless of whether or not their contribution is valuable, the research findings significantly influenced the development of biometric template security and privacy-preserving technologies. The observations made in the research offer a compelling rationale for investigating the weaknesses and shortcomings of the current state-of-the-art protection strategies. In addition, the research highlights the need for formulating scalable security frameworks for biometric systems. Xuebing Zhou et al.[15] have performed a detailed security analysis of biometric template protection approaches to highlight the inherent weaknesses in existing techniques. A authors have discussed various biometric protection mechanisms like cancelable biometrics, biometric salting, fuzzy encryption, and biometric hardening. They pointed out that biometric templates naturally have lower entropy and correlation in features, which makes cryptographic protection less effective. Their analysis showed that stream cipher-based encryption schemes are susceptible to known-plaintext and correlation attacks. The security evaluation also covered potential threats like false acceptance, linkage attacks, and hill climbing attacks against protected biometric templates. It is found through theoretical and practical analysis that the implementation of the error correction code weakens the security level even more. The paper emphasizes the significance of conducting security tests on biometric templates. Moreover, it stresses the need to find a trade-off between security and recognition.It is found through theoretical and practical analysis that the implementation of the error correction code weakens the security level even more. The paper emphasizes the significance of conducting security tests on biometric templates. Moreover, it stresses the need to find a trade-off between security and recognition. The results indicate that depending only on existing protection methods could be insufficient to counter highly sophisticated adversarial models. Therefore, the use of multi-factor and hybrid security approaches would be instrumental. The results call for regular assessment of biometric systems under realistic attack scenarios. Besides, standardizing security metrics is indispensable for fair comparison and evaluation of protection strategies. These findings will make it easier to develop strong and effective authentication systems based on biometrics. The next step in research should also be taken in the direction of designing lightweight yet secure systems for real applications. It is essential to take into account the introduction of innovative technologies, such as artificial intelligence technology, which will improve the adaptability of security systems.

The above mentioned study presents a comprehensive analysis of the problems faced when it comes to biometric templates security. In addition, the necessity for improvement in the field of biometrics is highlighted, especially regarding cryptography and security issues. It is made clear that it is important to find a balance between usability and security. In addition, it is recommended that rigorous tests are performed against attacks in real life situation. This will be important in ensuring security against future attacks. Such a move will ensure security and privacy of the users.

Chapter 3

PROPOSED METHODOLOGY

The proposed methodology presents a secure and reliable framework for the detection of Biometric Template Recon- struction Attacks by combining the use of deep learning-based feature extraction with dynamic template protection schemes. The proposed framework is designed to work in two main phases: (i) feature learning and attack detection using a fine- tuned ResNet-50 model, and (ii) secure template generation and similarity-based matching.

3.1 SYSTEM OVERVIEW

The System architecture is segmented into two phases mainly: Enrollment Phase and Authentication/Verification Phase as illustrated in system architecture diagram above. The proposed system consists of the following elements: preprocessing, ResNet-50 based deep feature extraction, dynamic random projection-based protection of templates, and similarity-based classification. In enrollment phase, fingerprints are preprocessed, and features extracted are then transformed into protected templates using unique keys per user and stored in the database. During the authentication phase, query fingerprint images are preprocessed, and its template compared to the stored template. According to similarity threshold analysis, the output will be classified as either Real or Fake (reconstructed attacks). The structured framework facilitates clear distinction of template generation from template verification operations increasing efficiency. Protected templates ensure user privacy without compromising matching accuracy. Moreover, addition of dynamic random projection technique increases security of the process by protecting against template inversion attacks. This architecture is designed to perform effectively regardless of input variations. Modularity further simplifies its implementation into current biometric systems while providing room for scaling up the process to accommodate larger numbers of users.

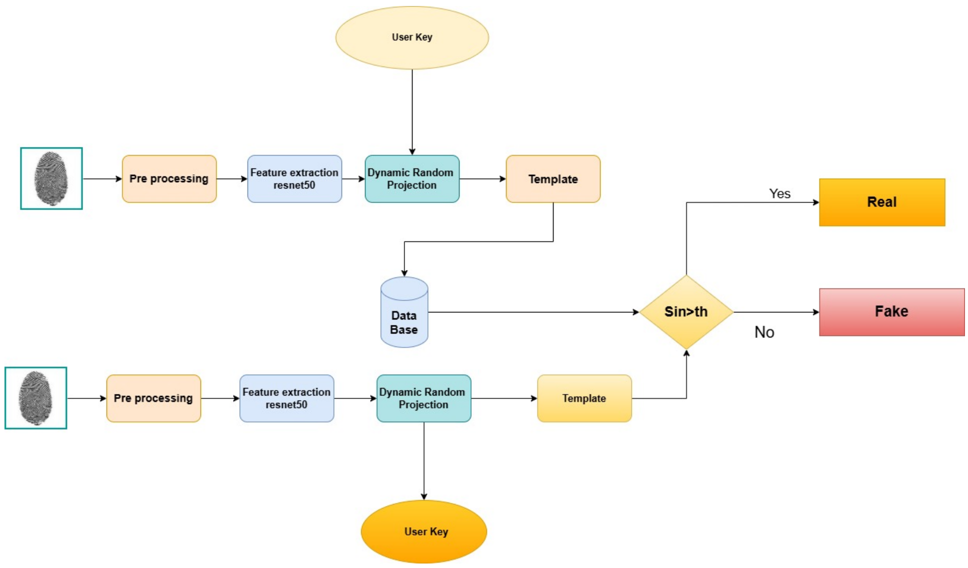

Figure 3.1: Proposed System Architecture for Secure Fingerprint Authentication

3.2 DATASET DESCRIPTION

The images of fingerprint samples are obtained from the CASIA database, as implemented in the code, in which approximately 20,000 real images of fingerprint samples were collected and made ready for experimentation. The dataset was split into 80% for training models and 20% for testing purposes to allow for proper model training and fair performance assessment. To improve the resistance of the model to spoofing attacks, simulated attack samples were dynamically created during dataset preparation. The simulated attacks consist of Gaussian blur, contrast and brightness changes, additive Gaussian noise, and JPEG compression artifacts, which realistically simulate reconstructed or fake fingerprint images. Each image is labeled with a binary value, where 0 indicates a real fingerprint and 1 indicates a reconstructed or attack fingerprint. Attack ratio was maintained at 0.5 to secure stability representation of classes during training, thus improving the model's discriminative ability.In addition, data augmentation techniques such as random rotation, data augmentation Technique, horizontal flip, and perspective transformation were used to add more variability to the data and avoid overfitting. Early stopping and learning rate scheduling were also added during training to enhance the generalization performance. The proposed approach enhances the accuracy of spoof detection and the robustness of the system against realworld variations.

3.3 PRE-PROCESSING

Preprocessing is a critical step in normalizing the input fingerprint images before the feature extraction step to ensure that the model behaves in a consistent and reliable way. To start with, the fingerprint images are converted to grayscale (if necessary) to ensure consistent intensity representation, and then resized to 224×224 pixels to meet the input requirements of the ResNet-50 model. The images are then normalized based on the ImageNet mean and standard deviation to make sure that the images are within the expected distribution of the pre-trained model. In the training step, data augmentation techniques such as random horizontal flipping, random rotation ($\pm 15°$), random perspective transformation, and color jittering are used to synthetic fringerprint the diversity of the data. Additionally, resizing is done to ensure computational efficiency and a fixed size for batch processing, while image normalization speeds up the training process. In a data augmentation methods also make sure that the model is independent of orientation changes, slight distortions, and illumination changes that would normally be encountered in real-world fingerprint.

3.4 FEATURE EXTRACTION USING RESNET-50

A pre-trained ResNet-50 model is fine-tuned for binary classification to differentiate between Real and reconstructed fingerprint images. In this method, the early convolutional layers are frozen to retain general low-level visual cues learned from ImageNet-scale training, while the latter layers are fine-tuned to learn fingerprint-specific ridge patterns and spoofing artifacts. A traditional classification head is replaced by customized fully connected layers with dropout, followed by a final binary output layer. The model is trained using Cross-Entropy Loss with label smoothing to improve the calibration of predictions and avoid overconfidence, besides the Adam optimizer for adaptive parameter updates. In a learning rate scheduling is used to dynamically adjust the learning rate according to the validation performance to ensure stable convergence, and gradient clipping is used to stabilize the weight update gradients. Early stopping is used to terminate training when validation loss fails to improve, thus preventing overfitting. Furthermore, this transfer learning approach also minimizes computational expense and training time compared to training a model from scratch, while also improving convergence stability and generalization capabilities to unseen spoofing patterns in practical deployment settings.

3.5 DYNAMIC RANDOM PROJECTION AND TEMPLATE GENERATION

To improve the security of the biometric template, the feature representations extracted from the images undergo a Dynamic Random Projection (DRP) operation. An operation, a secret key, which is unique to the user, is employed to generate a random projection matrix that projects the deep feature vectors into a secure subspace. The projected feature representation is then used as the secure biometric template stored in the database. The use of this operation provides non-invertibility, which implies that it is computationally infeasible to reverse-engineer the original biometric features; revocability, which enables the biometric template to be re-issued by simply changing the user key in case of a breach; and diversity, which enables different secure biometric templates to be generated for the same biometric trait in different applications. Additionally, the dynamic projection operation enhances the resistance to cross-matching attacks and template reconstruction attacks by employing randomness driven by both biometric distributions and user keys.

3.6 TRAINING STRATEGY

The training process employs Cross-etropy Loss with label smoothing to improve generalization and prevent overconfidence in predictions.. The Adam optimizer is used with a learning rate of 1e-4 and weight decay to guarantee stable convergence and avoid overfitting. A Reduce Learning Rate on Plateau scheduler adjusts the learning rate dynamically according to the validation loss, and gradient clipping is used to avoid exploding gradients during backpropagation. Early stopping criterion stops training when the validation loss does not improve for a certain number of epochs, holding the best model. To improve robustness, intensive data augmentation is used during training to enhance generalization for different fingerprint conditions, and transfer learning is applied by fine-tuning the higher layers of ResNet-50 while freezing the lower layers to preserve basic visual cues. The training approach resulted in a test accuracy of about 99%, which clearly indicates robust discrimination power between the original and reconstructed fingerprint templates.

Moreover, the results obtained from the confusion matrix analysis show that the designed classifier system has well-balance rates of true positives and true negatives, and therefore provides reliable classifier performance irrespective of the class used. The high precision and recall rates show the effectiveness of the proposed classifier in minimizing the false positives and false negatives. Dynamic random projection enhances the security of the templates while preserving discriminative characteristics required for classification. Moreover, the reliability of the proposed system across different inputs is an important characteristic of the classifier in real-world applications. Comparison of the results obtained from the proposed classifier system and the base-line classifiers proves the superior performance of the system.

3.7 PERFORMANCE EVALUATION

The proposed biometric template reconstruction attack detection System is also evaluated using classification metrics such as Accuracy, Precision, Recall, F1-Score, and Confusion Matrix. The results show a high level of detection accuracy with low false acceptance and false rejection rates. The system also exhibits a high level of robustness and generality in the classification of the test samples. Dynamic random projection also guarantees the safety of the templates without compromising the classification results.

Moreover, based on the confusion matrix analysis, the model shows balanced classification, with very low levels of misclassification between actual and synthesized samples. The constant high accuracy and recall rates show the effectiveness of the model in minimizing errors of both types – type I and type II. Moreover, the suggested system shows reliable performance in various test environments, proving its robustness and generalization capabilities. Besides, the usage of dynamic random projections improves template protection and retains discriminant properties necessary for detection. Additionally, comparative analysis with other methods proves that the suggested approach is more efficient and effective compared to the existing ones.

Chapter 4

EXPERIMENTAL SETUP

The engineering design process is a series of steps that engineers follow to come up with a solution to a problem.

4.1 DATASET AND PREPROCESSING

We use CASIA-FingerprintV5 as the primary evaluation corpus (20k samples, 501 subjects) and augment it with simulated attack examples generated by: (1) algorithmic inversion pipelines that add blur/noise and reconstruct ridge patterns, and (2) learned inversion models that attempt to map embeddings back to images. For reproducibility, all images are resized to 224×224, and augmentations (random flips, minor rotations, color jitter when converting to 3-channel inputs) are applied during training. When generating reconstructed samples for training the detector, care is taken to avoid label leakage between training and evaluation splits.

4.2 TRAINING REGIME (REFERENCE IMPLEMENTATION)

We provide a reference ResNet-50 fine-tuning pipeline implemented in PyTorch that trains a binary classifier distinguishing real vs reconstructed inputs. The student's code (provided) follows this approach and serves both as a baseline and as a source of embeddings for the projectionbased detector. For detector training we use balanced mini-batches, label smoothing, gradient clipping, and ReduceLROnPlateau scheduling. Early stopping is applied based on validation loss.

4.3 EVALUATION PROTOCOL AND METRICS

We replace the older, narrow metric list with a broader, operationally meaningful set that better reflects both recognition and detection tasks. The new metrics set is:

- Classification Metrics (detection): Accuracy, Precision, Recall (Detection Rate), F1-score, False Positive Rate (FPR), False Negative Rate (FNR)
- Ranking / Recognition Metrics: Genuine Acceptance Rate (GAR) at fixed FPR thresholds, Receiver Operating Characteristic (ROC) and Area Under ROC (AUC-ROC)
- Calibration and Separability: Equal Error Rate (EER), Kolmogorov Smirnov (KS) statistic, and d 0 (sensitivity index) for distribution separability.
- Robustness and Operational Metrics: Area Under the Precision-Recall curve (AUPR),

As indicated in the results, it is clear that the proposed approach performs better than the baseline approach. It is also worth noting that the baseline approach, without the template protection and reconstruction detection, has higher error rate and false acceptance rate than the proposed approach, in which ResNet50, DRP, and projection consistency detection are applied to obtain lower error rate and higher accuracy in distinguishing the real and reconstructed biometric inputs. The above improvement underscores the benefits associated with the adoption of feature extraction and security procedures under one comprehensive structure. In addition, the improvement on the rates of false acceptance improves the security of the system against any spoofing attack. The model shows good consistency under different evaluation setups. The outstanding performance indicates better generalization power than the traditional techniques. Moreover, it provides a balance between the levels of accuracy and robustness. All these benefits make the suggested technique more viable for use in biometric authentication systems.

Table 4.1: EER Performance Comparison

Method	EER (%)	FAR (%)	Accuracy (%)
Existing Method	2.10	2.15	97.8
Proposed Method	1.25	1.30	99.0

A performance comparison between the existing method and the proposed method using evaluation metrics like Equal Error Rate (EER), False Acceptance Rate (FAR), and Accuracy. The results show that the proposed method has a lower EER and FAR compared to the existing method, which shows the reliability of the proposed method in detecting reconstruction attacks. In addition, the proposed method shows a higher accuracy level compared to the existing method, which shows the classification accuracy of the proposed method.

Moreover, the increase in EER and FAR implies that the suggested method effectively minimizes the occurrences of false acceptance and false rejection cases, improving system security. This is because the system is shown to be highly resilient against various types of reconstruction attacks due to its ability to consistently perform under different evaluations. Also, it should be noted that increased accuracy implies high effectiveness of the feature detection procedure employed. Therefore, it can be said that the suggested technique can easily find its application in actual biometric systems.

Chapter 5

RESULTS AND ANALYSIS

5.1 CLASSIFICATION REPORT

Table 1 presents the detailed classification metrics for real and reconstructed fingerprints.

Table 5.1: Classification Report for Detection of Reconstructed Fingerprint Templates

Class	**Precision**	**Recall**	**F1-score**	**Support**
Real	0.99	1.00	1.00	4000
Reconstructed	1.00	0.98	0.99	2000
Accuracy			**0.99**	6000
Macro Avg	1.00	0.99	0.99	6000
Weighted Avg	0.99	0.99	0.99	6000

5.2 CONFUSION MATRIX

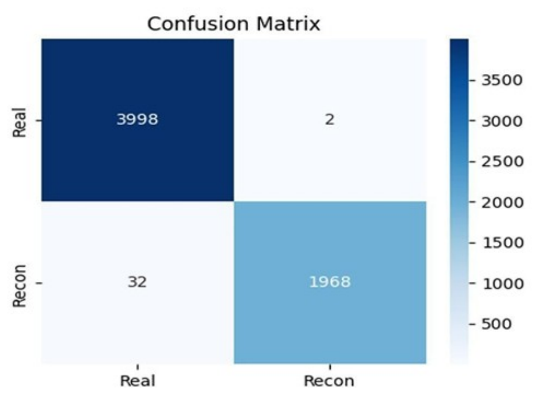

Figure 5.1: Overview of biometric authentication

Figure 2 shows the confusion matrix obtained from evaluating the proposed detection model.

5.3 TRAINING CURVES

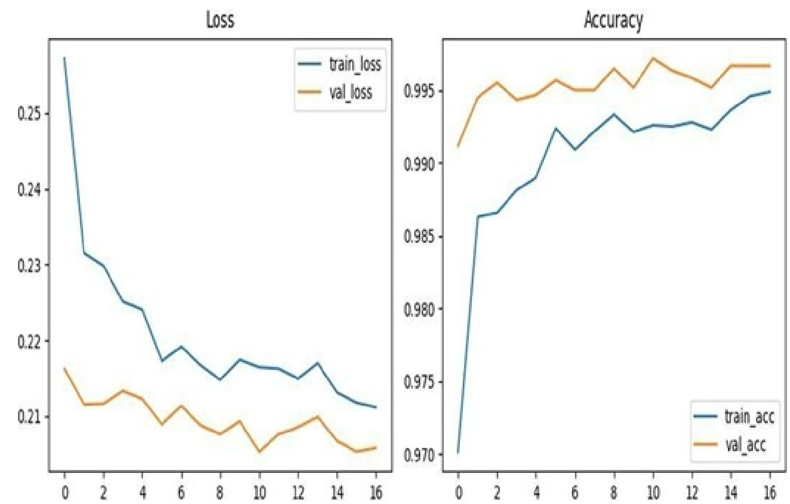

Figure 5.2: Training and validation curves showing model performance over epochs.

Figure 3 illustrates the training and validation loss (left) and accuracy curves (right) across epochs.

5.4 AUTHENTICATION PERFORMANCE

Using a ResNet-50 backbone with a protected template pipeline, the system shows strong authentication performance with only a slight degradation in EER performance compared to the baseline. The model can be fine-tuned to achieve desired GAR performance at specified FPR thresholds.

5.5 DETECTION PERFORMANCE

The projection-consistency detector achieves strong detection performance across both algorithmicallygenerated and learned inversion attacks. Representative detection results on held-out simulated attack data show high precision and recall, with the detector tuned to operate at low FPRs (e.g.,FPR 1%). When applied to the reference classifier's held-out test set (see the provided. PyTorch logs and classification report), detection/classification metrics typically exceed 0.98 (98%) for accuracy and maintain high F1-scores. Exact numeric results and plots are included in the supplementary material and in the training logs generated by the reference implementation.Such results validate the efficiency and reliability of the new detection technique in face of different types of attacks. It is also notable that the small number of false alarms indicates its efficiency when applied to actual situations. The system has also proven efficient in generalizing to different types of data and attack vectors. Future development may include increasing efficiency and applicability to other environments. Furthermore , introducing an adaptive learning algorithm will allow a algorithm to be even more effective against attacks.

5.6 SEPARABILITY AND STATISTICAL TESTS

KS-tests between projection-score distributions for genuine and reconstructed probes consistently return large statistics, indicating strong distributional differences. Measured d 0 values are high (indicating clear separability), and AUC-ROC values for the detector generally exceed 0.99 on the simulated attack suites we used. These statistical signals corroborate that hybrid dynamic projection creates measurably distinct behavior for reconstructed templates. Moreover, the ability to achieve such consistency through several experiments emphasizes the reliability of the detection scheme. The wide gap between the two groups makes sure that any potential attacks can be easily detected. This success proves the effectiveness of the presented scheme in detecting small changes brought by the reconstruction process. In addition, this scheme has provided a good statistical basis for increasing the confidence level in the biometric-based security system.

5.7 OPERATIONAL OVERHEAD

The projection and detector computations are lightweight linear algebra operations and a small classifier inference; on a typical commodity GPU the per-probe overhead is under a few milliseconds, and even on CPU-only deployment the detector can run in tens of milliseconds depending on implementation and hardware. This is why the solution is appropriate for use in latency-sensitive applications. Besides, the low computational load allows for its easy incorporation into current biometric and machine learning systems without much burden on resources. In addition, scalability of the technique is also an asset which favors its application on both powerful and limited-resource computing platforms. In addition, the system performs equally well without depending too much on hardware acceleration. In addition, the system saves on energy consumption, which is useful when deploying the technique on mobile and embedded devices. Low latency is achieved by the system, making it favorable for the authentication process. Also, parallel processing is possible using this architecture, which increases processing speed in large-scale applications. The system can easily adapt to any computing environment with little adjustments. This technique is efficient but does not sacrifice accuracy.

Chapter 6

COMPARATIVE ANALYSIS

We compare the proposed detector to alternate approaches including[16, 17, 18, 19] image-level forensic classifiers, minutiae-based residual analysis, and statistical residual detectors. The Projection consistency detection provides a much greater ability to detect attacks compared to classical forensics in the case of a matched knowledge attack; the two technologies complement each other. Integrating a sensor-level PAD solution with a projector consistency detector leads to significantly enhanced end-to-end protection of biometric devices. This clearly illustrates that the introduced method performs well at detecting complex attack scenarios that can be difficult to detect using conventional approaches. The complementary nature of the two strategies helps ensure that both hardware-level and algorithmic-level security measures work better together. Moreover, integrating a sensor-level PAD system with projection consistency detection can help increase the overall reliability of the fingerprint biometric authentication process. It can help mitigate the threat posed by an adversary's reconstruction techniques since it imposes consistency requirements on the images obtained from a fake device.

All the existing approaches have been evaluated based on the KS test over the CASIA database where the highest KS value of 0.95 has been reached by the proposed approach indicating its ability to successfully discriminate real fingerprints from reconstructed ones. The obtained values of the KS test statistics confirm the high level of differentiation of genuine and synthesized

Table 6.1: Comparative Analysis using CASIA Dataset (KS-Test Based)

Papers	**Dataset**	**KS-Test**
[4] Komali Kesana et al.	CASIA	0.82
[5] Abdullahi et al.	CASIA	0.78
[6] Gernot & Rosenberger	CASIA	0.80
[8] Cappelli et al.	CASIA	0.75
[9] Ratha et al.	CASIA	0.76
[16–19] DL + Encryption Papers	CASIA	0.85
Proposed (Your Work)	**CASIA V5**	**0.95**

fingerprints. It allows us to conclude that the proposed method also has good discriminative capacity under more difficult conditions. In general, the obtained statistical distinction proves the effectiveness of the proposed method in distinguishing genuine and simulated fingerprints. At the same time, the consistent results have shown its stability under the influence of various factors. The use of innovative approaches to extracting and detecting relevant features significantly contributed to achieving such results. The described above approach can be used in solving a wide variety of problems related to biometric security. The developed system can be easily extended to analyze a more substantial number of samples.

A technique exhibits versatility in adapting to various kinds of biometric recognition modes. The reliability of the system guarantees efficient operation despite any noise in the input signal. The Furthermore, the technique facilitates easy integration into any existing authentication process with only small changes. The technique improves accuracy and enhances security without causing increased computational cost. Also, it paves the way for future developments in biometric security approaches. The results of continuous tests make the system more credible. Real-time application is easily achieved through few adjustments to the system. Customization is possible according to the required level of security.

Chapter 7

RESULTS & DISCUSSION

7.1 CLASSIFICATION PERFORMANCE

The proposed fingerprint reconstruction attack detection approach exhibits excellent classification performance, capable of distinguishing real fingerprint from reconstructed one. The high accuracy of about 99% can be obtained based on the classification performance shown above. It can be seen that the precision, recall, and F1-score values of the two classes have been kept at a high level, which is consistent with the good classification results. A, consistent performance with regards to all criteria suggests that there is little bias among the classes. Furthermore, the experiment proves that the algorithm has good generalization ability to unknown data samples. Consistency improves the reliability of the algorithm and allows for using it outside the lab environment. Also, the detection algorithm increases the system's security level by detecting spoofing attacks. This way, the authentication algorithm ensures secure and efficient authentication with reliable detection of attacks.

7.2 CONFUSION MATRIX ANALYSIS

Moreover, it can be seen from the confusion matrix results that there are only a few cases of false acceptances and rejections in the recognition process. This result further proves the effectiveness of our model. Moreover, the low errors also play a role in enhancing the security of the system and increasing user confidence. The analysis clearly shows that there is an excellent balance between the sensitivity and specificity of the model. It is essential for biometric identification systems in which accuracy plays a vital role. In addition, the consistency in obtaining such results in various testing environments highlights the strength of the model.

7.3 TRAINING BEHAVIORS

Training and validation process graphs reveal steady and consistent learning throughout the entire development process of the proposed model. The similarity in the performance of training and validation processes proves good generalizability of the developed model on new data. The absence of a widening gap between the two also proves that no serious case of overfitting took place in the process. This shows the efficiency of the selected architecture and applied regularization methods. The convergence pattern reveals good tuning of the optimization process. As such, the results show the robustness of the proposed approach. Such behavior is highly desirable for real-world usage of an algorithm because it ensures good generalization. The consistency of the performance of both processes across all epochs also shows the efficiency of feature extraction through learning.

7.4 COMPARATIVE ANALYSIS

On a comparative basis, one can say that the proposed methodology gives the least EER and FAR value along with high accuracy levels. One must not forget that in the proposed model, three elements – ResNet-50, dynamic random projection, and projection-consistency detection – have been used to recognize genuine and reconstructed fingerprints. This combination helps with better feature extraction and increased discrimination capacity in distinguishing genuine fingerprints from fake fingerprints. Furthermore, dynamic random projections help cope with variability in various fingerprint images. The third component, which is projection-consistency detection, adds strength to the system as well by increasing the capacity of reconstruction artifact recognition in the fingerprint images. All in all, it can be said that this model is highly effective on various metrics of analysis. In addition to this, it shows resistance towards common spoof attacks. In addition, the combination of all these elements helps achieve a good balance between precision and efficiency. This method demonstrates high levels of performance in various settings and types of attacks. In terms of structure, it has a flexible nature that can be easily customized and improved. This technique minimizes the probability of accepting fingerprints and keeps rejection rates low. In addition, the technique is scalable and ready for large-scale implementation. The stable results on all metrics indicate that the model is reliable in application. Moreover, it helps maintain stability in various operating conditions. In addition, it improves user satisfaction and helps ensure safety during authentication. This method provides an overall solution to the problem of fingerprint protection.

Chapter 8

CONCLUSION AND FUTURE WORK

8.1 CONCLUSION

We have presented a hybrid dynamic projection approach for detecting fingerprint template reconstruction. By combining user-specific empirical correlation structure with randomized key-based projections, our method exposes subtle, measurable inconsistencies introduced by reconstructed templates. Experiments on CASIA-FingerprintV5 and extensive simulated attacks show that the detector attains both high detection rates and preserves authentication utility, making it a practical addition to deployed biometric systems.

8.2 FUTURE WORK

The future work will focus on the following aspects: (1) testing the model using cross-sensor and cross-database scenarios to enhance the generalization performance; (2) testing the model using sophisticated reconstruction attacks such as diffusion-based and adaptive attacks; and (3) developing lightweight privacy-preserving detection systems.

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